CSULB
CECS174

Variables, Types and Arithmetic operation

Objectives:
At the end of this lab, you should be able to
1. Identify legal/illegal variable name
2. Select the most suitable data type
3. Write a simple python code with input / output and basic arithmetic operations

1- Circle whether the following variables are legal or illegal. If you circle illegal explain why.

1. Legal / Illegal NUM_STUDENTS
2. Legal / Illegal 2Students This is illegal because the variable start with a number
3. Legal / Illegal int
4. Legal / Illegal Tampa! This is illegal because the variable end with a special character
5. Legal / Illegal firstName
6. Legal / Illegal zip code This is illegal because there is space between within the variable
7. Legal / Illegal num2
8. Legal / Illegal _middle_initial_ This is illegal because the variable start with a special character
9. Legal / Illegal #three This is illegal because the variable end with a special character
10. Legal / Illegal tree

2- Circle the most appropriate data type given the description.

1. int / float / string Number of students in CECS174 class.
2. int / float / string A student's address.
3. int / float / string Due date.
4. int / float / string An hourly pay rate.
5. int / float / string The current month.
6. int / float / string Table's length.
7. int / float / string The speed of your car (miles per hour).
8. int / float / string Social Security Number (SSN).
9. int / float / string A driver license Number.
10. int / float / string Dewey Decimal Classification

3- Write a Python code that works as a calculator. Your code prompts the user to input two integer values, then calculates and displays the results of the addition, subtraction, multiplication, division, integer division, modulo, and exponent of those two numbers.

Demo your finished program to your lab instructor as instructed during the lab.
Example Output (user's input is in italics):

Please enter two values:
Value #1 = *3*
Value #2 = *5*

3 + 5 = 8
3 - 5 = -2
3 * 5 = 15
3 / 5 = 0.60
3 // 5 = 0
3 % 5 = 3
3 ^ 5 = 243

Code:

```
num1 = int(input("enter the first number: "))
num2 = int(input("enter the second number: "))
operator = input("enter operator: ")

if(operator == '+'):
    resultNum = num1 + num2
elif(operator == '-'):
    resultNum = num1 - num2
elif(operator == '*'):
    resultNum = num1 * num2
elif(operator == '/'):
    resultNum = num1/ num2
elif(operator == '// '):
    resultNum = num1 // num2
elif(operator == '%'):
    resultNum = num1 % num2
elif(operator == '^'):
    resultNum = num1 ^ num2

print(resultNum)
```